Answer Key & Teacher Notes

Session 1 — Finding the Mean · Session 2 — Reaching a Target Mean

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Finding the Mean

TALK PROMPTS

- Why is the mean described as a fair share of the data?
- Why must the mean of 7, 4, 9, 6, 9 (Problem 1b) land between 4 and 9?

PARTNER TASK

Above the Mean. One partner finds the mean of 8, 12, 9, 11, 15 (Problem 1a); the other names the values above the mean. Switch with 20, 16, 22, 18, 25, 25 (Problem 1c).

WATCH FOR

- Reporting the total as the mean (forgetting to divide by the count).
- Leaving out zeros, which divides by too few values.
- Dividing by the greatest value instead of the number of values.

SESSION 2 · TEACHER NOTES

Reaching a Target Mean

TALK PROMPTS

- Why do you work with the total instead of with the separate scores?
- Ravi (Problem 3) needs 110 in game 5 for a mean of 102. What happens to the score he needs if he lowers his target mean to 101?

PARTNER TASK

Swim Meet Swap. In Problem 2, one partner finds the target total (29 × 5 = 145); the other finds the meet-five score (33). Switch with Problem 4: mean 22; values 19, 25, 21, 28 (fifth number 17).

WATCH FOR

- Multiplying by the old count instead of the new count that includes the missing value.
- Subtracting the target mean from the sum so far instead of building a target total.
- Thinking one high score pulls the mean all the way up to that score.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.SP.B.5c	1	2	1	2	1	1	2	2	2	1	1
S1 · Form B	6.SP.B.5c	1	2	1	2	1	1	2	2	2	1	1
S2 · Form A	6.SP.B.5c	1	2	2	1	2	2	2	2	2	1	1
S2 · Form B	6.SP.B.5c	1	2	2	1	2	2	2	2	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

mean: The fair share: add all values, then **divide** by how many there are.

average: Another **name** for the mean.

measure of center: One number that tells a **typical** value of the data.

data set: All the **values** you collected.

- Mean = **sum** of the values ÷ **number** of values.
- Count every value, even a **zero**.
- The mean lands between the **least** and greatest values.

Example: Add: 3 + 5 + 6 + 10 = **24** Count the values: there are **4**. Divide: 24 ÷ 4 = **6**. Check: 6 is between 3 and 10. Answer: The mean is **6** stickers.

YOU TRY

1. a. 35 ÷ 5 = 7 b. 15 ÷ 3 = 5

2. 100 ÷ 5 = $20

EXIT TICKET

1. a. 68 ÷ 4 = 17 b. 72 ÷ 6 = 12

2. 410 ÷ 5 = 82 points

3. **C** (average)

PRACTICE FORM A

1. a. 55 ÷ 5 = 11 b. 35 ÷ 5 = 7 c. 126 ÷ 6 = 21 d. 180 ÷ 6 = 30

2. a. 9 + 11 + 14 + 16 + 16 + 18 = 84 visits b. 84 ÷ 6 = 14 visits

3. 60 ÷ 5 = 12 cm

4. No. 70 ÷ 4 = 17.5; the mean uses every value, including 40.

5. 28 ÷ 4 = 7 stickers (this is the mean)

6. 95 ÷ 5 = $19

7. No. The $0 day is a value, so divide by 5: 50 ÷ 5 = $10 (not 50 ÷ 4 = $12.50).

8. a. 240 ÷ 6 = $40 b. Rise; 96 > 40. New mean: 336 ÷ 7 = $48

9. 36 is the total, not the mean. She must divide by the number of values: 36 ÷ 4 = 9.

10. **A** (9)

11. **B** (17)

Self-check word **AVERAGE**

PRACTICE FORM B

1. a. 45 ÷ 5 = 9 b. 40 ÷ 5 = 8 c. 120 ÷ 6 = 20 d. 150 ÷ 6 = 25

2. a. 8 + 10 + 12 + 13 + 15 + 20 = 78 visits b. 78 ÷ 6 = 13 visits

3. 70 ÷ 5 = 14 cm

4. No. 120 ÷ 4 = 30; the mean uses every value, including 60.

5. 32 ÷ 4 = 8 stickers (this is the mean)

6. 105 ÷ 5 = $21

7. No. The $0 day is a value, so divide by 5: 55 ÷ 5 = $11 (not 55 ÷ 4 = $13.75).

8. a. 252 ÷ 6 = $42 b. Rise; 98 > 42. New mean: 350 ÷ 7 = $50

9. 40 is the total, not the mean. She must divide by the number of values: 40 ÷ 4 = 10.

10. **C** (5)

11. **B** (number of values)

Self-check word **BALANCE**

GUIDED NOTES

target mean: The mean you **want** to reach.

target total: Target mean × number of **values**.

missing value: Target total − the sum you **already** have.

- Work **backward**: target total = target mean × number of values.
- Then **subtract** the sum so far to find the missing value.

Example: Count all the scores, including the missing one: **4**. Target total: 8 × 4 = **32**. Sum so far: 6 + 7 + 9 = 22. Subtract: 32 − 22 = **10**. Answer: The fourth score is **10**. Check: 32 ÷ 4 = 8

YOU TRY

1. a. 100 b. 15

2. 90 × 4 = 360; 85 + 90 + 88 = 263; 360 − 263 = 97 points

EXIT TICKET

1. a. 54 b. 20

2. 80 × 5 = 400; 70 + 85 + 78 + 81 = 314; 400 − 314 = 86 points

3. **A** (number of values)

PRACTICE FORM A

1. a. 80 b. 56 c. 15 d. 24

2. a. 29 × 5 = 145 b. 27 + 31 + 24 + 30 = 112 c. 145 − 112 = 33 points

3. 102 × 5 = 510; 96 + 104 + 88 + 112 = 400; 510 − 400 = 110

4. 22 × 5 = 110; 19 + 25 + 21 + 28 = 93; 110 − 93 = 17

5. Dana is wrong. 70 × 4 = 280; 280 + 100 = 380; 380 ÷ 5 = 76

6. 45 × 6 = 270; 41 + 39 + 50 + 47 + 44 = 221; 270 − 221 = 49°F

7. 90 × 4 = 360; 70 + 74 + 78 = 222; 360 − 222 = 138. Not possible: 138 > 100.

8. a. 18 × 4 = 72; 72 − 51 = 21 kg b. 20 × 4 = 80; 80 − 51 = 29 kg

9. He subtracted the mean instead of building a target total. Target total = 30 × 5 = 150; 150 − 104 = 46 points.

10. **C** (80)

11. **A** (9)

Self-check word **TARGETS**

PRACTICE FORM B

1. a. 90 b. 72 c. 18 d. 28

2. a. 30 × 5 = 150 b. 26 + 33 + 28 + 31 = 118 c. 150 − 118 = 32 points

3. 104 × 5 = 520; 98 + 106 + 90 + 110 = 404; 520 − 404 = 116

4. 24 × 5 = 120; 20 + 27 + 22 + 29 = 98; 120 − 98 = 22

5. Marco is wrong. 75 × 4 = 300; 300 + 100 = 400; 400 ÷ 5 = 80

6. 52 × 6 = 312; 48 + 55 + 50 + 57 + 51 = 261; 312 − 261 = 51°F

7. 92 × 4 = 368; 72 + 76 + 80 = 228; 368 − 228 = 140. Not possible: 140 > 100.

8. a. 20 × 4 = 80; 80 − 54 = 26 kg b. 22 × 4 = 88; 88 − 54 = 34 kg

9. She subtracted the mean instead of building a target total. Target total = 32 × 5 = 160; 160 − 112 = 48 points.

10. **B** (40)

11. **A** (target total)

Self-check word **MISSING**

SHEET 1 · FINDING THE MEAN

1. a. 9 b. 25 c. 5 d. 5 e. 3.5 f. 22
2. a. 25 goals b. 10 games c. 25 ÷ 10 = 2.5 goals per game
3. Sum = 3.0 in; 3.0 ÷ 5 = 0.6 in
4. It goes up: 20 is greater than the old mean. New mean = (8 + 10 + 12 + 20) ÷ 4 = 50 ÷ 4 = 12.5
5. Zero is a data value, so there are 4 values, not 3. Mean = 24 ÷ 4 = 6 (not 8).

SHEET 2 · REACHING A TARGET MEAN

1. A: total 36, missing 11 B: mean 16, missing 15
2. 110 × 5 = 550; 112 + 98 + 105 + 121 = 436; 550 − 436 = 114 points
3. $50 × 5 = $250; $45 + $60 + $52 + $38 = $195; $250 − $195 = $55
4. a. 91 ÷ 7 = 13 push-ups b. 14 × 8 = 112; 112 − 91 = 21 push-ups
5. She must multiply by 4, the number of tests including the new one: 85 × 4 = 340. 340 − 246 = 94, so she needs a 94.
6. **B** (11)
7. **C** (70)

October 2026 conceptual additions — teacher key

Session 1 · Finding the Mean

Total 16 cans, four groups; mean 4 cans per group. The zero contributes nothing to the total but is still one group sharing the total. A fair-share model redistributes 6 from the group with 10: 4 to the zero group and 2 to the group with 2.

Extension: No. Groups with 1 and 2 cans have mean 1.5, which neither group has.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Reaching a Target Mean

For mean 8, target total is 4 × 8 = 32; current total 23; missing score 9. For mean 9, target total 36; missing score 13, impossible with a maximum of 10.

Extension: (23 + 10) ÷ 4 = 8.25. No larger fourth score is allowed.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.